Kinematic Modelling and MATLAB-Based Analysis of the Dobot Magician Lite Robot Arm

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Figure 1: Dobot Magician Lite, taken from [4].

ABSTRACT

This project presents the kinematic modelling and MATLAB-based analysis of the Dobot Magician Lite (DM-Lite) robot arm with a MATLAB implementation of the model.

1. INTRODUCTION

Robot manipulators are essential in industrial, medical and educational domains because of their precision, repeatability and programmability [2,11]. Common applications include pick-and-place, assembly, and laboratory automation. In educational environments, arms such as the DM-Lite are used for teaching robotics.

Classified as an anthropomorphic manipulator [11], its all-revolute configuration offers superior dexterity and a large spherical workspace relative to its footprint. however,

it lacks direct correspondence between joint DOFs and Cartesian space, resulting in variable positioning accuracy and significant gravitational stress compared to Cartesian or SCARA systems.

This project develops a MATLAB-based model to evaluate its kinematics, Jacobian, singularities, and dynamic behavior.

2. KINEMATIC MODELLING

2.1 Robot configuration and assumptions

The kinematic model treats the robot as a 4R serial manipulator, with θ_1 describing base rotation, θ_2 and θ_3 defining planar arm motion, and θ_4 controlling tool orientation.

2.1.1 The Passive Joint

The physical robot is a series of 4-bar linkages [1].

The passive joint P is introduced to maintain the constraint that the end-effector remains parallel to the ground. This is achieved by defining the passive joint angle as $-\theta_3 - \theta_2$.

The final rotation into the end effector frame α_3 happens at the end of the passive link, so the passive link's rotation still applies to the frame from joint 3 as shown in Table 1.

2.2 Denavit-Hartenberg Parameterisation

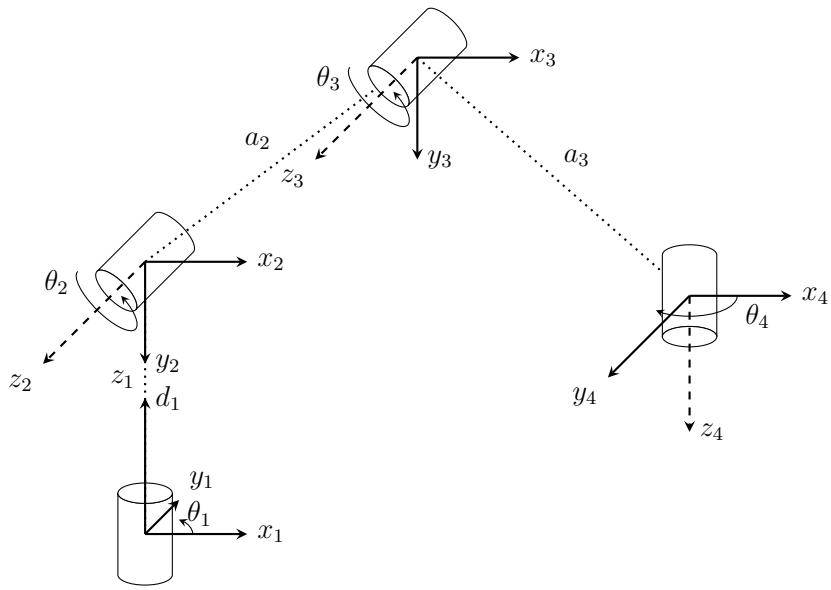

Figure 2: Denavit-Hartenberg frame assignment used for the kinematic modelling of the DM-Lite manipulator. z_2 and z_3 are coming out of the page.

The coordinate frames were assigned according to the standard Denavit-Hartenberg convention [11]. Each z-axis was aligned with the axis of rotation of the corresponding

joint, while each x-axis was defined along the common normal between adjacent joint axes.

The Denavit-Hartenberg parameter table derived from the frame assignment is

Link	a_i **(mm)**	α_i	d_i **(mm)**	θ_i
1 - Base	0	90°	107	θ_1
2 - Rear Arm	150	0°	0	θ_2
3 - Fore Arm	150	0°	0	θ_3
P - Passive	0	90°	0	$-\theta_3 - \theta_2$
4 - End effector	0	0°	0	θ_4

Table 1: Denavit-Hartenberg parameter table.

2.3 Forward kinematics

The forward kinematics of the DM-Lite were derived using the Denavit-Hartenberg convention. Homogeneous transformation matrices were derived for each joint and multiplied to create the transformation matrix from the base frame to the end-effector frame, providing both rotation and cartesian position [2, 3].

Let r be the radial distance from the base z-axis to the wrist center:

$$r = L_2 \cos(\theta_2) + L_3 \cos(\theta_2 + \theta_3) \tag{1}$$

The Cartesian position of the end-effector is then:

$$x = r \cos(\theta_1) \tag{2}$$
$$y = r \sin(\theta_1) \tag{3}$$
$$z = L_1 + L_2 \sin(\theta_2) + L_3 \sin(\theta_2 + \theta_3) \tag{4}$$

2.4 Jacobian and singularity analysis

The Jacobian matrix relates joint rates to end-effector motion. In this work, the geometric Jacobian was used to describe the relationship between joint velocities and the linear and angular velocity of the end-effector, while an analytical form was considered for singularity analysis in terms of the selected task variables [11,12]. Jacobian-based analysis is particularly useful for identifying configurations in which the manipulator loses mobility or becomes difficult to control.

Geometric Jacobian

The geometric Jacobian relates the joint velocities to the linear and angular velocity of the end effector:

$$\begin{bmatrix} v \\ \omega \end{bmatrix} = J_g(q)\dot{q}$$

The geometric Jacobian is therefore:

$$J_g(q) = \begin{bmatrix} -r\sin(\theta_1) & \cos(\theta_1)(-L_2\sin(\theta_2) - L_3\sin(\theta_2+\theta_3)) & -L_3\sin(\theta_2+\theta_3)\cos(\theta_1) & 0 \\ r\cos(\theta_1) & \sin(\theta_1)(-L_2\sin(\theta_2) - L_3\sin(\theta_2+\theta_3)) & -L_3\sin(\theta_2+\theta_3)\sin(\theta_1) & 0 \\ 0 & L_2\cos(\theta_2) + L_3\cos(\theta_2+\theta_3) & L_3\cos(\theta_2+\theta_3) & 0 \\ 0 & -\sin(\theta_1) & -\sin(\theta_1) & 0 \\ 0 & \cos(\theta_1) & \cos(\theta_1) & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix}$$

Singularities

A singularity occurs when the determinant of the analytical Jacobian becomes zero:

$$\det(J) = L_2 L_3 \sin(\theta_3) r = 0$$

This expression shows that singularities occur when either $\sin(\theta_3) = 0$ or $r = 0$. The condition $\sin(\theta_3) = 0$ corresponds to the arm being fully stretched or fully folded, so that the links become collinear. The condition $r = 0$ corresponds to the wrist lying on the base axis, where the manipulator loses motion authority in certain horizontal directions. Near these configurations, the velocity mapping becomes ill-conditioned and large joint velocities may be required to generate small end-effector motions [11, 12].

2.5 Inverse kinematics

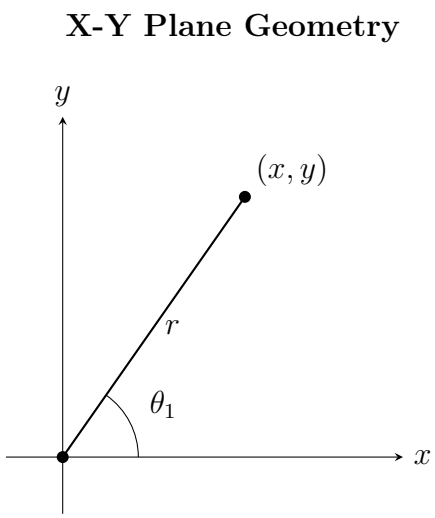

Figure 3: Top-view geometry showing the base joint angle θ_1 relative to the Cartesian coordinates (x, y).

The base joint angle is:

$$\theta_1 = \text{atan2}(y, x)$$

The remaining problem is reduced to a planar two-link geometry in the r–z plane [2]. Define $z' = z - L_1$ as the height relative to the shoulder. The function $\text{atan2}(y, x)$ is used

because it returns the correct angular quadrant from the Cartesian coordinates [10].

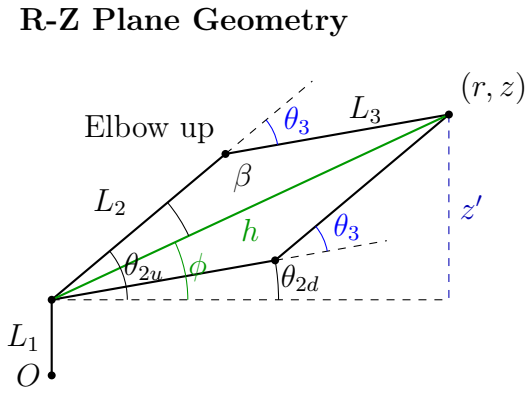

Figure 4: r–z plane geometry used for inverse kinematics of the shoulder and elbow joints.

Two solution branches exist for reachable targets: an elbow-up configuration and an elbow-down configuration. Using the law of cosines in the triangle formed by L_2, L_3, and the distance to the target $h = \sqrt{r^2 + (z')^2}$:

$$\cos\beta = \frac{L_2^2 + h^2 - L_3^2}{2L_2 h} \tag{5}$$

The shoulder angles are then:

$$\theta_{2u} = \phi + \beta \quad \text{and} \quad \theta_{2d} = \phi - \beta$$

where $\phi = \text{atan2}(z', r)$ is the angle from the horizontal to the target.

For θ_3, the forearm angle, we have two cases as well: **elbow-up** and **elbow-down**. However, the derivation for the inverse kinematic equation for this angle is a bit more complex.

Given

$$r^2 = x^2 + y^2 \tag{6}$$

$$z' = z - L_1 \tag{7}$$

$$r^2 = L_2 cos(\theta_2) + L_3 cos(\theta_2 + \theta_3) \tag{8}$$

$$z' = L_2 sin(\theta_2) + L_3 sin(\theta_2 + \theta_3) \tag{9}$$

Adding r^2 and z'^2 and simplifying gives us

$$\cos\theta_3 = \frac{r^2 + z'^2 - L_2^2 - L_3^2}{2L_2 L_3} \tag{10}$$

$$\sin\theta_3 = \sqrt{1 - \cos^2\theta_3} \tag{11}$$

And the value of $\sin\theta_3$ above is negative for the elbow-up condition

$$\theta_{3d} = \text{atan2}(\sin\theta_3, \cos\theta_3) \quad \text{(for elbow-down)} \tag{12}$$

$$\theta_{3u} = \text{atan2}(-\sin\theta_3, \cos\theta_3) \quad \text{(for elbow-up)} \tag{13}$$

3. MATERIALS AND METHODS

3.1 Kinematic Modelling Approach

The kinematic model of the DM-Lite was developed using the Denavit–Hartenberg (DH) convention. Each joint was modelled as a revolute joint, and the robot was approximated as an open kinematic chain, despite the physical presence of a four-bar linkage mechanism.

3.2 Inverse Kinematics Strategy

A geometric approach was selected for the inverse kinematics problem, leveraging the direct geometric relationships between joint angles and end-effector positions.

This approach was preferred over numerical methods due to its computational efficiency, analytical clarity, and suitability for real-time applications [2, 9].

3.3 Dynamic Modelling Approach

The dynamic behaviour of the manipulator was considered using the standard rigid-body robot dynamics formulation [11, 12]

$$\tau = M(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q)$$

where τ is the vector of joint torques, $M(q)$ is the inertia matrix, $C(q, \dot{q})\dot{q}$ represents Coriolis and centrifugal effects, and $G(q)$ is the gravity vector.

In the present project, the dynamic equations were incorporated at a conceptual and modelling level to extend the kinematic model developed in MATLAB. However, a complete high-fidelity dynamic implementation was limited by the lack of detailed physical parameters such as link masses, mass centres, and inertia tensors from the manufacturer documentation [11]. Therefore, the work focused on establishing the dynamic form of the equations and explaining how such terms would be integrated into the MATLAB model for future extension.

3.4 MATLAB Implementation

The robot model was implemented as described in Section 2 as a MATLAB class with an API allowing for moving the joint angles, setting the end effect location in 3D space and viewing the transformation matrices, jacobian etc. See dobot.m [5] for the full implementation.

A UI [8] was built (see Figure 5) to allow users to interact with the model and unit tests (dobot_tests.m [6]) were added to ensure the model stayed accurate over time. A separate supplementary document, Mathematical Derivations.docx [7], is included in the submission ZIP and contains the full step-by-step derivations that were omitted from the main report for brevity.

3.5 AI

AI tools, specifically Gemini & ChatGPT, were used to assist in generating MATLAB code for the UI and unit tests. The dobot.m [5] MATLAB class was written by hand as that contains the mathematical model.

4. RESULTS AND VALIDATION

The MATLAB model was validated through forward kinematics (FK), inverse kinematics (IK), and singularity testing. FK was verified by comparing computed end-effector positions against geometric expectations, while IK was validated by reconstructing joint configurations from target Cartesian coordinates. Consistency between these mappings confirms the model's internal mathematical integrity.

Furthermore, tests for unreachable targets and singularity detection [6] demonstrate that the implementation correctly handles workspace edge cases. The results show strong agreement between the analytical derivations and the MATLAB implementation.

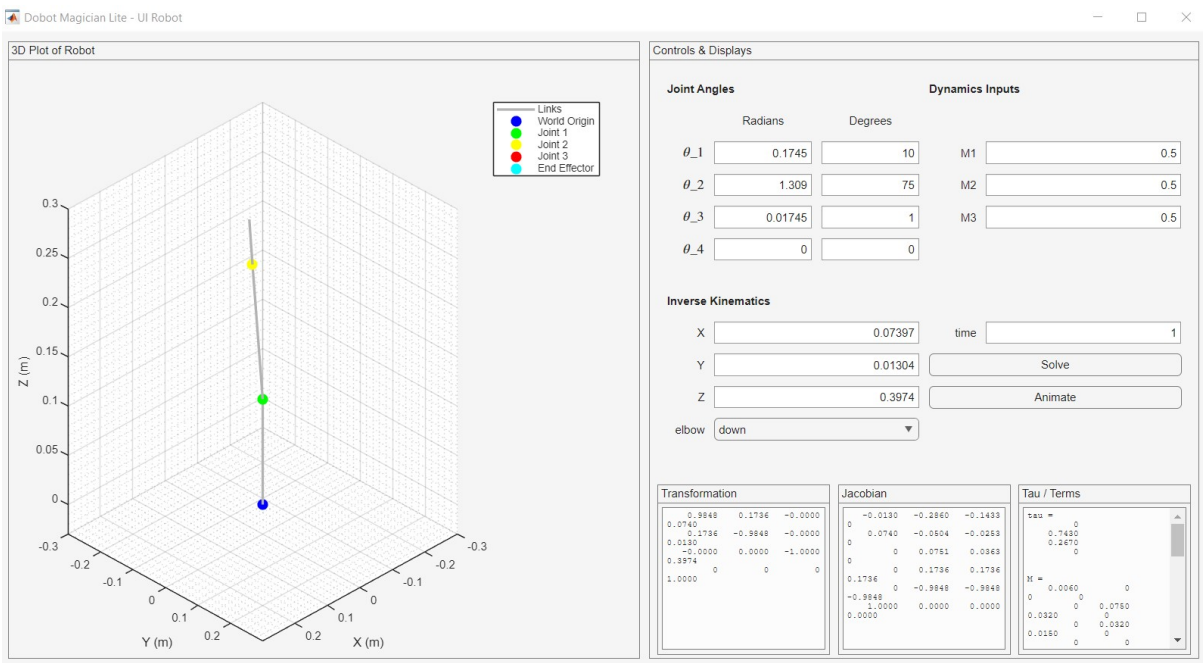

Figure 5: Successful inverse kinematics solution using the MATLAB-based UI.

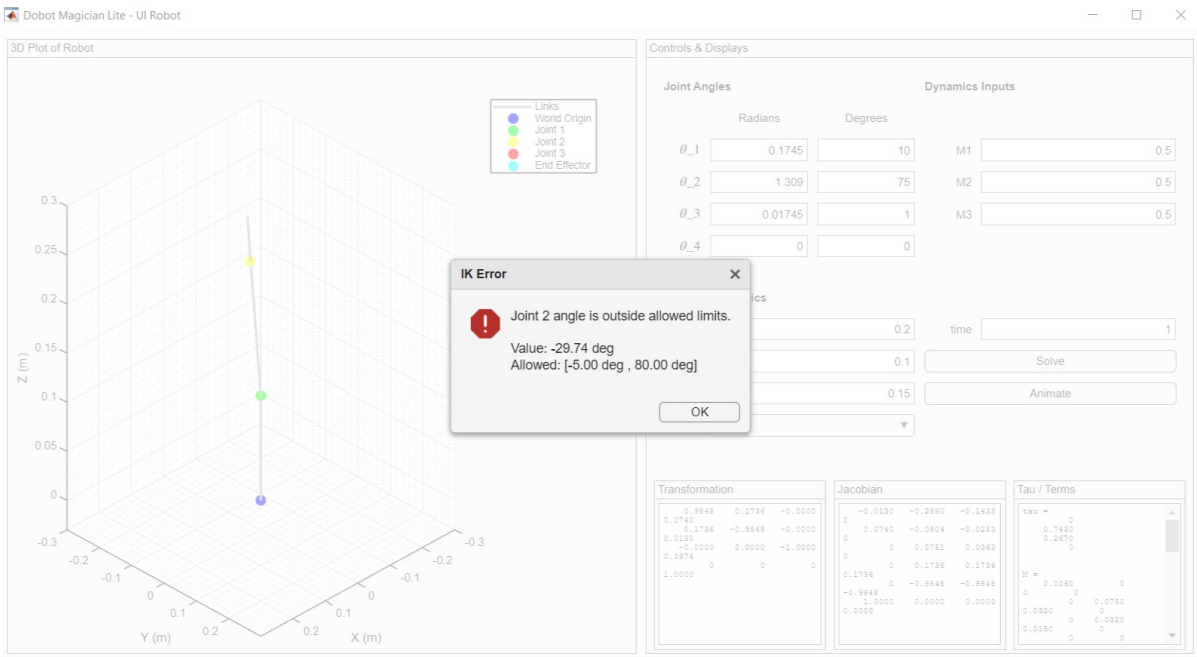

Figure 6: Detection of unreachable target in inverse kinematics.

5. DISCUSSION AND CONCLUSION

This project developed and validated a kinematic model of the DM-Lite using the Denavit–Hartenberg formulation. Forward and inverse kinematics, the Jacobian for differential motion, and singular configurations were derived analytically and implemented in MATLAB. The resulting model successfully reproduced valid end-effector positions while rejecting targets outside the reachable workspace.

A primary challenge was the robot's 4-bar linkage system; simplifying this to a serial chain initially caused the end-effector to lose its ground-parallel orientation, a problem addressed in Subsection 2.1.1.

Another issue with the Inverse kinematics was solved by reliably computing solutions reducing the 3D problem to a planar geometric construction in the r–z plane. Due to limited manufacturer data, link masses, centers of mass, and inertia tensors were estimated.

While the current MATLAB UI utilizes a cosine easing function for joint-space animation, future work will employ the Jacobian to enable Cartesian path-following.

6. REFERENCES

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